

PS-2: Development of a Collocated SAR-GPM IMERG Dataset for Rainfall Characterization in Synthetic Aperture Radar Observations

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Problem Statement

Synthetic Aperture Radar (SAR) observations are widely used for ocean surface monitoring; however, the presence of rain significantly alters backscatter signatures.

The satellite-based precipitation products like GPM IMERG provide reliable rain rate estimates at high temporal resolution that can be used to study the rain characteristics in SAR backscatter images.

The challenge is to develop a framework for collocating SAR observations with GPM-IMERG rain rate data and generate a comprehensive dataset to analyze and characterize the impact of rainfall on SAR backscatter.

Datasets used

1. GPM- IMERG :-https://disc.gsfc.nasa.gov/datasets/GPM_3IMERGHH_07/summary
2. Sentinel 1 SAR data: <https://browser.dataspace.copernicus.eu/>

Abstract

Rainfall significantly influences microwave radar backscatter measured by Synthetic Aperture Radar (SAR), particularly over ocean surfaces.

Understanding this relationship is important for improving interpretation of SAR imagery and developing rainfall-sensitive remote sensing applications.

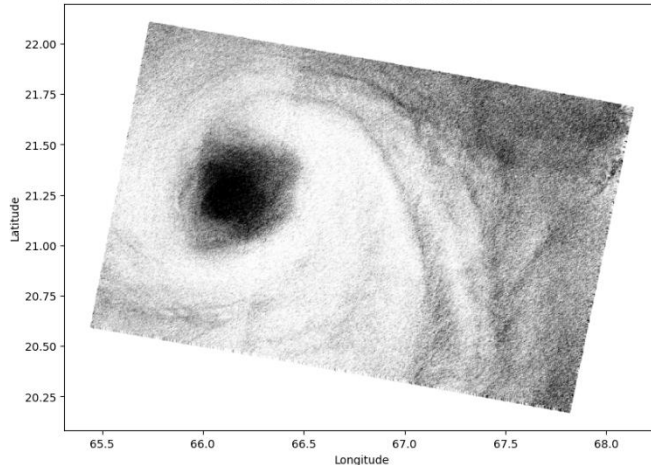
This project develops a collocated dataset linking Sentinel-1 SAR observations with precipitation measurements from the Global Precipitation Measurement (GPM) Integrated Multi-satellite Retrievals for GPM (IMERG) product.

A nearest-neighbor spatial collocation methodology is implemented to associate each sampled SAR pixel with the corresponding IMERG rainfall estimate.

The resulting dataset contains radar backscatter measurements, geographic coordinates, incidence angle information, and rain-rate observations. Statistical analysis demonstrates measurable rainfall-related changes in SAR backscatter, especially within the cross-polarized VH channel. The developed dataset provides a foundation for future rainfall characterization and machine learning studies using SAR observations.

STUDY DATA

Sentinel-1 VH Backscatter

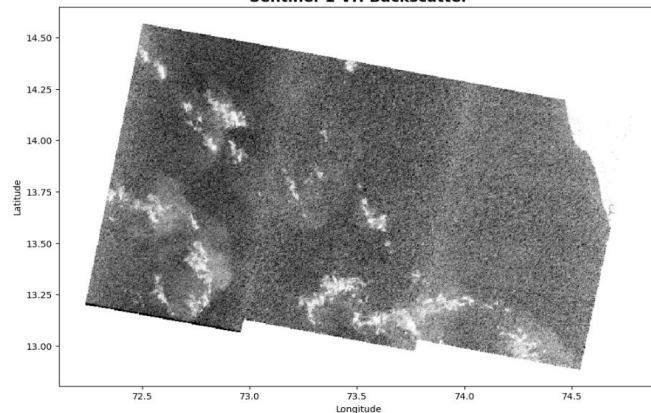


June 14, 2023

Cyclone Biparjoy

The cyclone in arabian sea can be a very good dataset to analyze the impact of very heavy rainfall with high wind conditions on SAR backscatter.

Sentinel-1 VH Backscatter

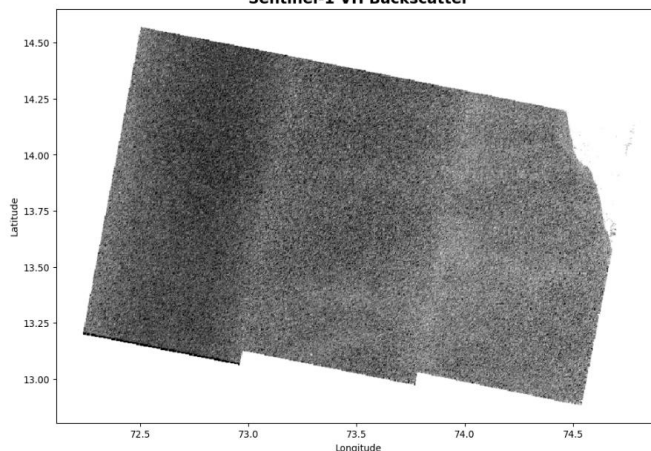


May 14, 2025

Moderate Rain near Goa,

1-20 mm per hour rain range, near coast of goa with little land contamination to study the impact of rainfall on SAR images.

Sentinel-1 VH Backscatter



January 2, 2025

Exact same place,

Exact same place under dry/no rain conditions to compare the backscatter with rainy conditions as in image above it.

Sentinel-1 SAR Data

Sentinel-1 is a C-band SAR mission operated by the European Space Agency (ESA). We used IW swath GRD data from it. The mission provides dual-polarization observations suitable for ocean monitoring.

The following variables were extracted from the SAR scene:

- Latitude
- Longitude
- VV backscatter
- VH backscatter
- Incidence angle

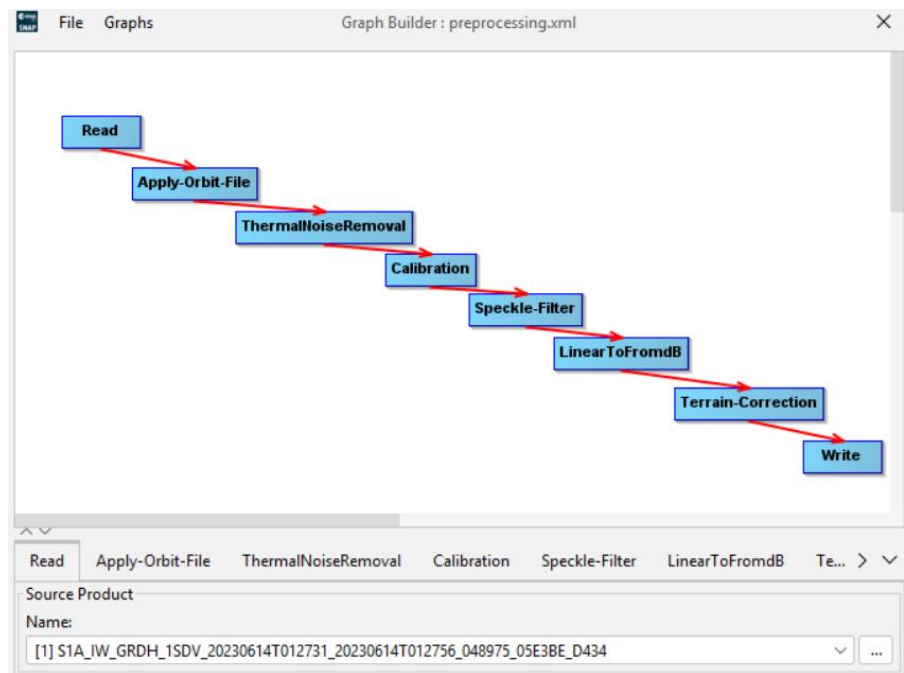
The SAR data

was preprocessed

using ESA SNAP and

exported to NetCDF format

for further analysis.



**** Drive-link with Preprocessing Graph, Code and Collocated Datasets: - ****

https://drive.google.com/drive/folders/14MtnhgDXPwLFuYA7hLYIJUeMJrJh8k5O?usp=drive_link

1. **Thermal Noise Removal:** Sentinel-1 GRD products contain thermal noise that manifests as systematic intensity banding, particularly at subswath boundaries. This step corrects for that noise to improve radiometric consistency across the scene.
2. **Radiometric Calibration:** The raw pixel values in a SAR image are arbitrary digital numbers (DN). Calibration converts them into physically meaningful backscatter coefficient values (Sigma0), enabling comparison across different scenes, acquisition dates, and sensors.
3. **Speckle Filtering:** SAR imagery is inherently affected by speckle, a multiplicative noise arising from coherent signal interference. Speckle appears as a granular salt-and-pepper texture that complicates visual interpretation and degrades

segmentation model performance. Filtering reduces this effect while preserving spatial features.

4. **Terrain Correction:** Although GRD products are already detected and projected, terrain correction maps the image precisely to a standard geographic coordinate system using a Digital Elevation Model (DEM), correcting for remaining geometric distortions caused by sensor geometry and topographic relief.

IMERG Processing

The IMERG HDF5 product was opened and the precipitation variable was extracted. Latitude and longitude grids were used to construct a spatial reference system for collocation.

Temporal Collocation

The temporal collocation of the images was done manually; we looked into the timestamps for acquisition time of the SAR images and downloaded the nearest half hour range global precipitation data from GPM-IMERG dataset.

Spatial Collocation

A KDTREE nearest-neighbor approach was used for efficient spatial matching.

For every SAR pixel:

- Latitude and longitude were identified.
- The nearest IMERG grid cell was located.
- Rainfall rate from the matched IMERG cell was assigned.

This procedure produced one rainfall value for every sampled SAR pixel observation.

Choice of Data Formats

The processed Sentinel-1 SAR observations were stored in the NetCDF (NC) format after preprocessing in SNAP. NetCDF was selected because it preserves multidimensional geospatial data structures, associated metadata, coordinate information, and multiple raster bands within a single self-describing file. Compared to GeoTIFF, NetCDF provides improved reliability for handling large datasets, facilitates efficient access through Python scientific libraries, and allows direct integration with array-oriented processing workflows.

The final collocated SAR-IMERG datasets were stored in the Apache Parquet format. Parquet is a compressed, columnar storage format optimized for analytical applications involving large tabular datasets. It offers significant reductions in storage requirements, faster read and write performance, and seamless compatibility with data analysis libraries

such as Pandas. Since the developed framework generates datasets, whose sizes vary with scene dimensions and sampling strategies, Parquet provides an efficient and scalable solution for storing collocated observations.

Software Libraries and Tools

The implementation of the proposed framework primarily relied on open-source Python libraries together with ESA's SNAP toolbox.

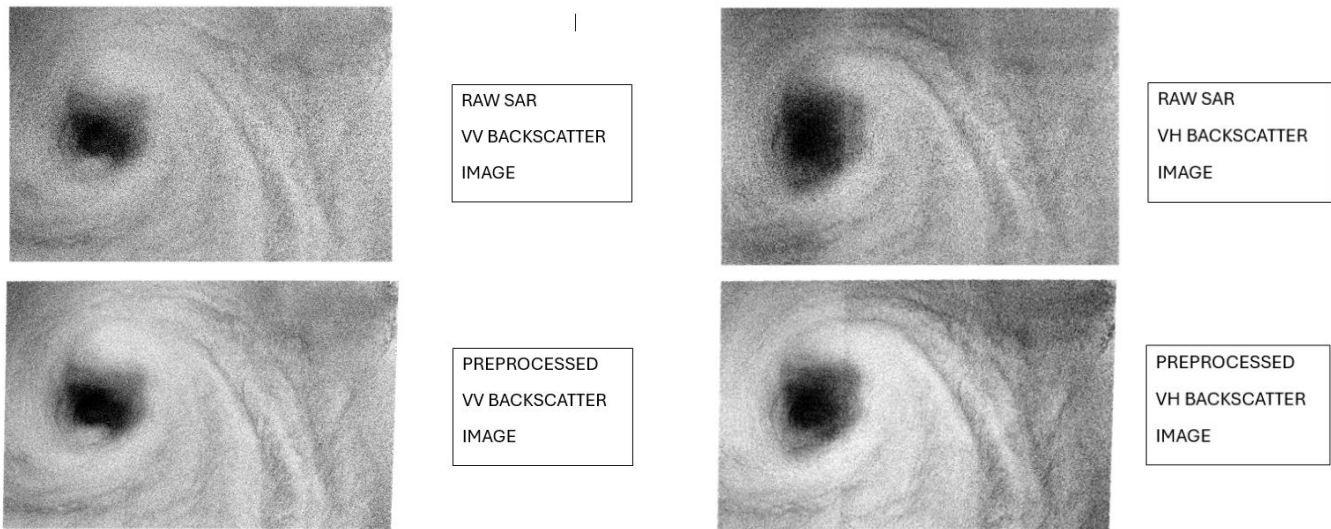
Software Libraries Used in the Project:-

Library/Tool	Purpose
SNAP	Sentinel-1 preprocessing and graph-based processing
Python	Primary programming language
NumPy	Numerical computations and array operations
Pandas	Tabular data handling and statistical analysis
Xarray	Reading and processing multidimensional NetCDF datasets
h5py	Accessing IMERG HDF5 precipitation products
SciPy	KDTree implementation and scientific computations
Matplotlib	Visualization of backscatter and rainfall relationships
PyArrow	Reading and writing Parquet datasets
Shapely	Geometric operations for spatial filtering
GeoPandas	Vector-based geospatial analysis
Natural Earth Dataset	Land-sea masking and coastline information
Jupyter Notebook	Interactive development and experimentation environment

The use of these libraries enabled the construction of a modular and reproducible processing framework capable of handling Sentinel-1 SAR observations, GPM IMERG precipitation products, spatial collocation procedures, statistical analyses, and visualization tasks. The framework was designed to remain flexible so that additional scenes, alternative precipitation products, or machine learning techniques can be incorporated with minimal modification to the existing workflow.

A stride-based subsampling approach was employed to reduce the computational burden associated with processing high-resolution Sentinel-1 SAR imagery. Rather than considering every SAR pixel, pixels were sampled at regular intervals using a stride value of 40, thereby significantly decreasing processing time, memory requirements, and dataset size while maintaining a representative spatial distribution of observations. The stride parameter can be adjusted according to the desired balance between computational efficiency and spatial sampling density.

A Comparison of preprocessed and raw images in different polarizations



Workflow of the Collocation Task

The complete workflow consists of:

1. SAR data loading.
2. IMERG rainfall loading.
3. Spatial downsampling of SAR pixels.
4. Coordinate extraction.
5. Spatial collocation.
6. Dataset generation.
7. Statistical analysis.

Invalid observations were removed

using the following criteria:

- Missing VV values removed.
- Missing VH values removed.
- Missing incidence angles removed.
- Non-physical rainfall values removed.

This ensured that only physically meaningful observations were retained for analysis.

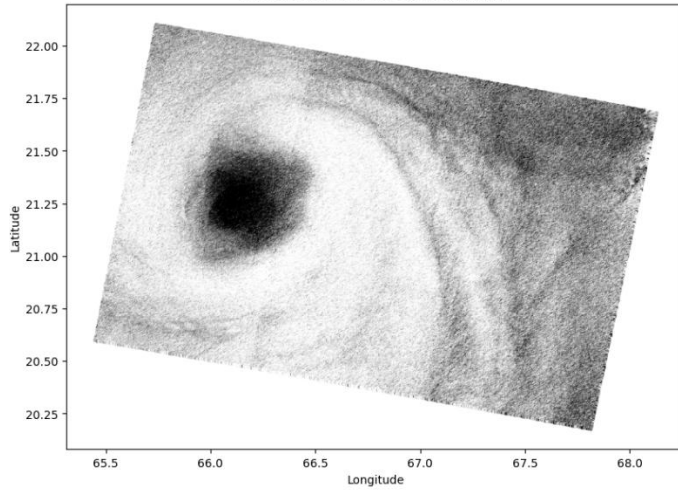
The collocated dataset structure

The final dataset is in apache parquet format with the following columns in tabular format:

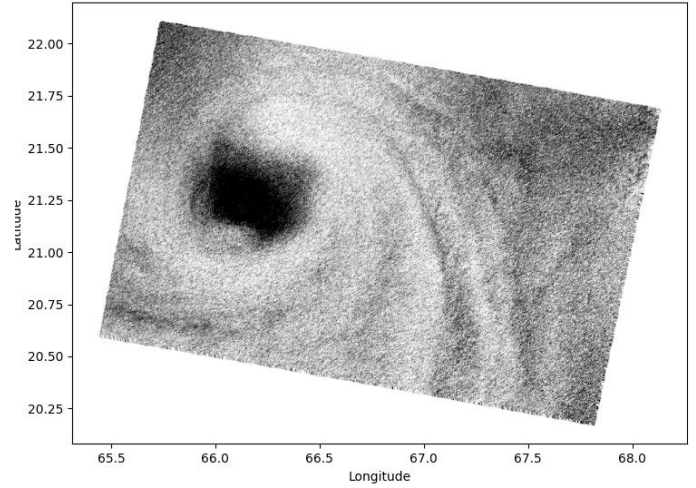
“lat” “lon” “VV_dB” “VH_dB” “Incidence_angle” “rain_mm_hr”

The IMERG spatial resolution is coarser than SAR resolution. That’s why the rainfall data appears as a grid on the SAR image. Owing to the coarse spatial resolution of IMERG, fine-scale precipitation structures detectable in SAR imagery may not be fully resolved, introducing uncertainties in pixel-level collocation.

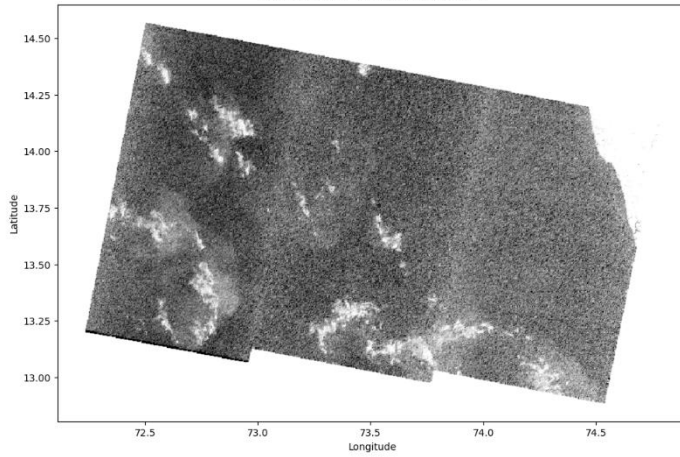
Sentinel-1 VH Backscatter



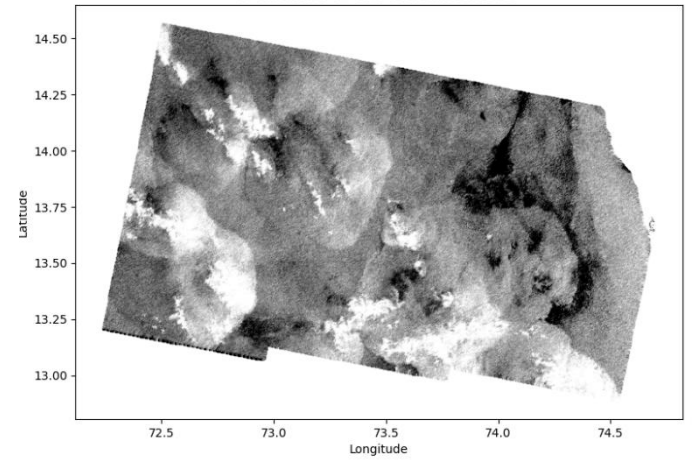
Sentinel-1 VV Backscatter



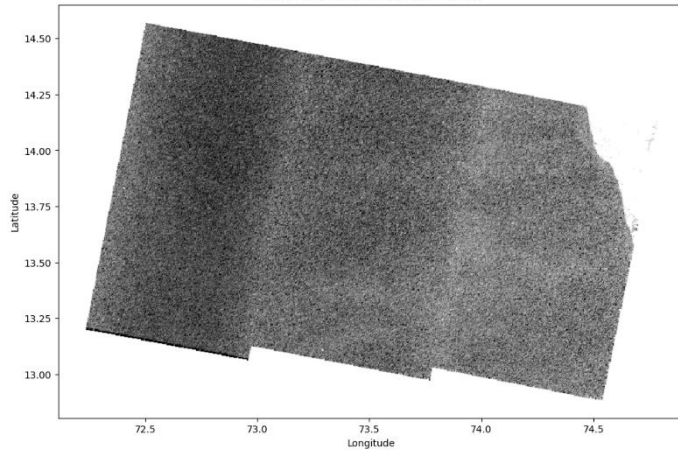
Sentinel-1 VH Backscatter



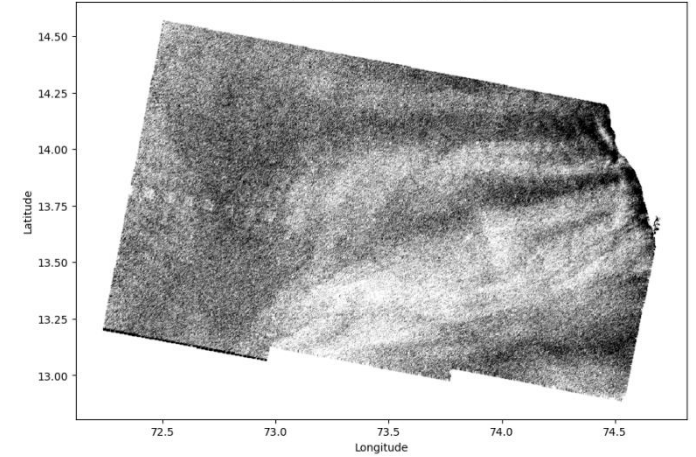
Sentinel-1 VV Backscatter



Sentinel-1 VH Backscatter



Sentinel-1 VV Backscatter



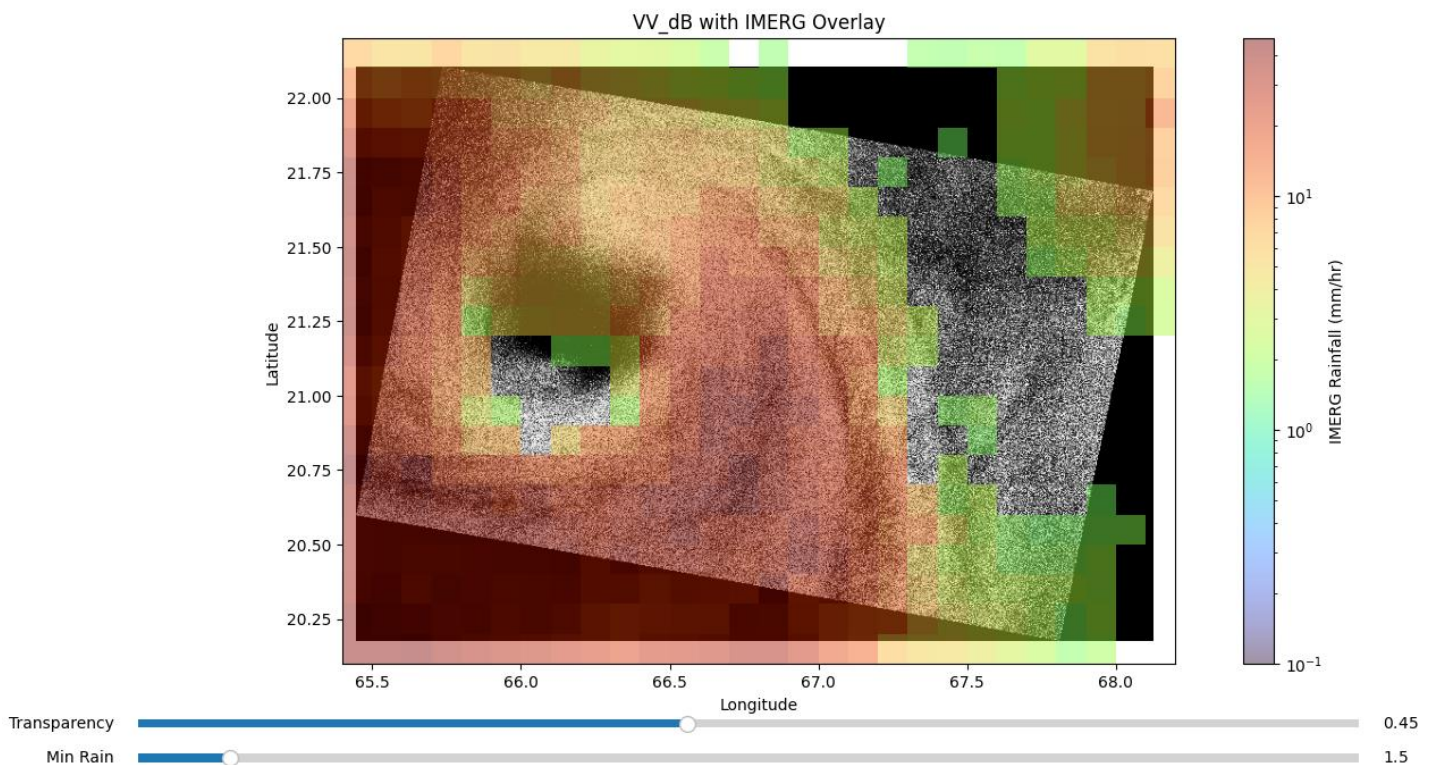
#VH vs VV backscatter Images

#Land–Sea Masking

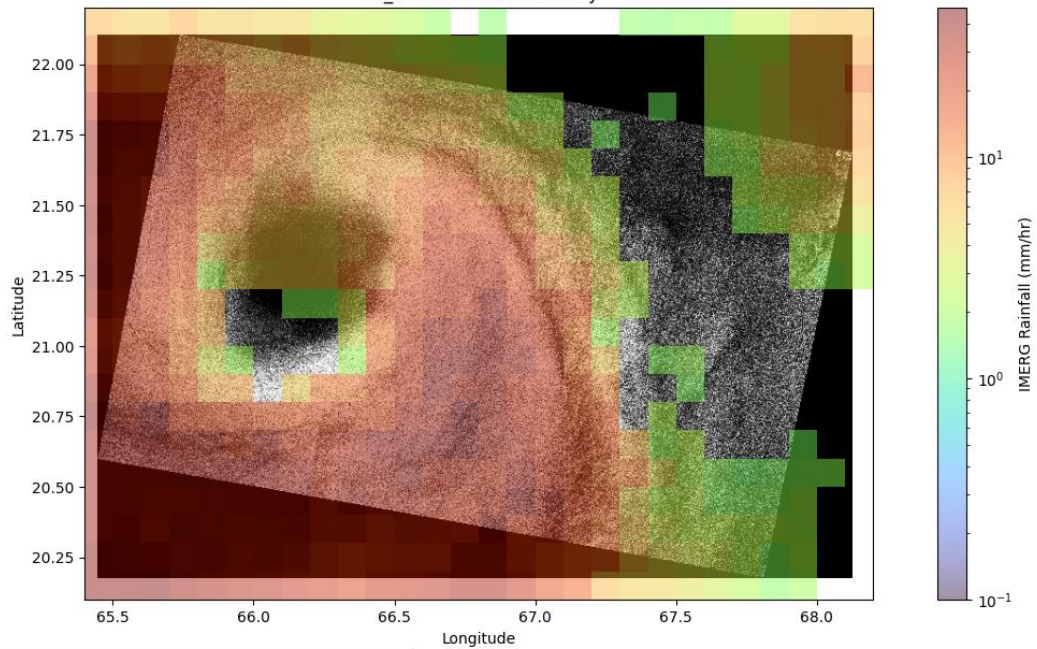
To ensure that the analysis focused solely on oceanic rainfall signatures, observations corresponding to land surfaces were removed from the collocated dataset using a land–sea mask derived from the Natural Earth coastline dataset. This filtering step minimizes contamination from terrestrial backscatter characteristics, which differ significantly from ocean surface scattering mechanisms, thereby improving the reliability of the rainfall–SAR relationship analysis.

IMERG–SAR Overlay Visualization

To facilitate visual inspection of rainfall signatures in SAR observations, IMERG precipitation estimates were overlaid onto the geocoded Sentinel-1 SAR imagery. The overlay was generated using geographic coordinates, ensuring that only precipitation pixels falling within the spatial extent of the SAR scene were displayed. Adjustable transparency was employed to enhance the simultaneous interpretation of SAR backscatter patterns and collocated rainfall distributions, providing a qualitative assessment of the spatial correspondence between observed precipitation and SAR-derived features.

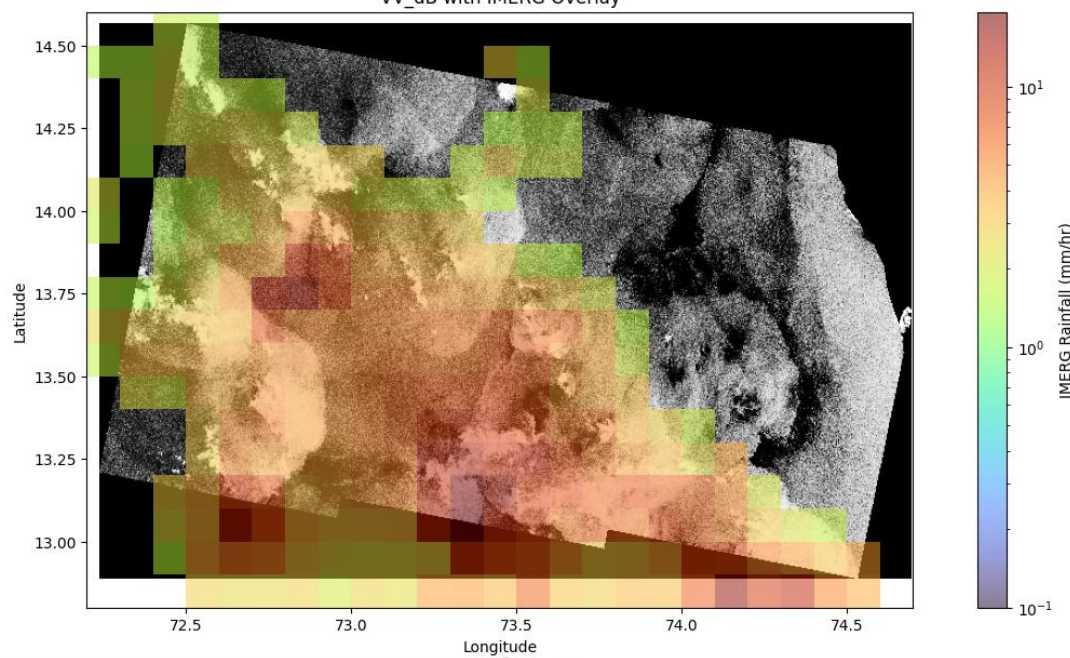


VH_dB with IMERG Overlay

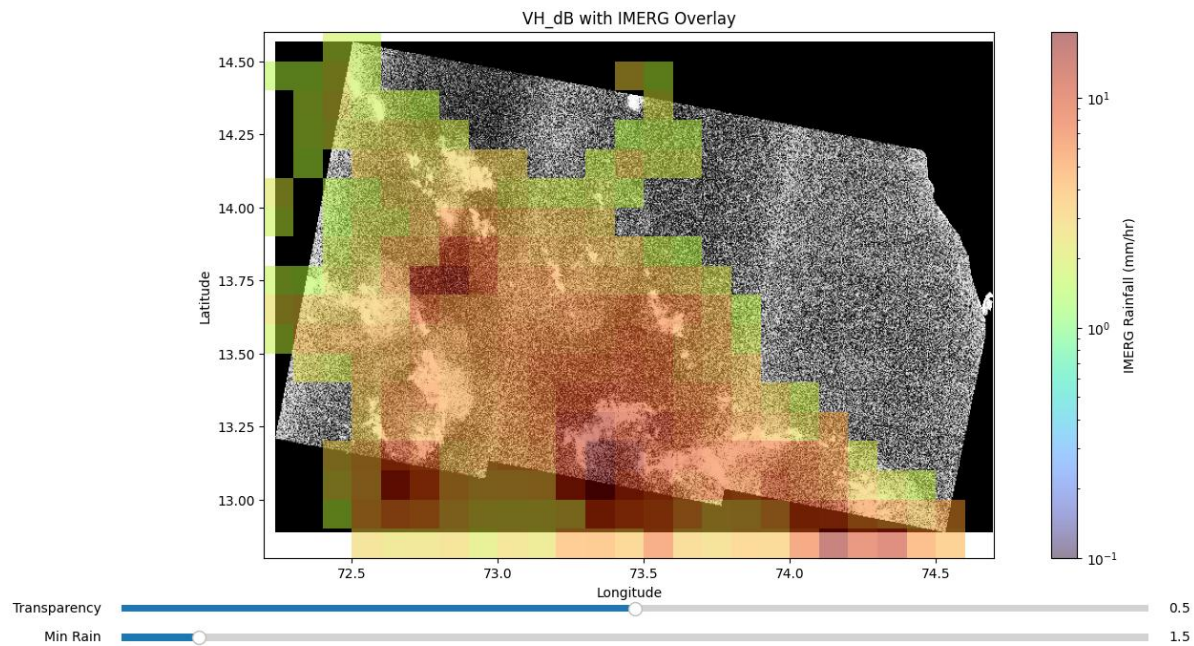


Transparency 0.45
Min Rain 1.5

VV_dB with IMERG Overlay



Transparency 0.5
Min Rain 1.5

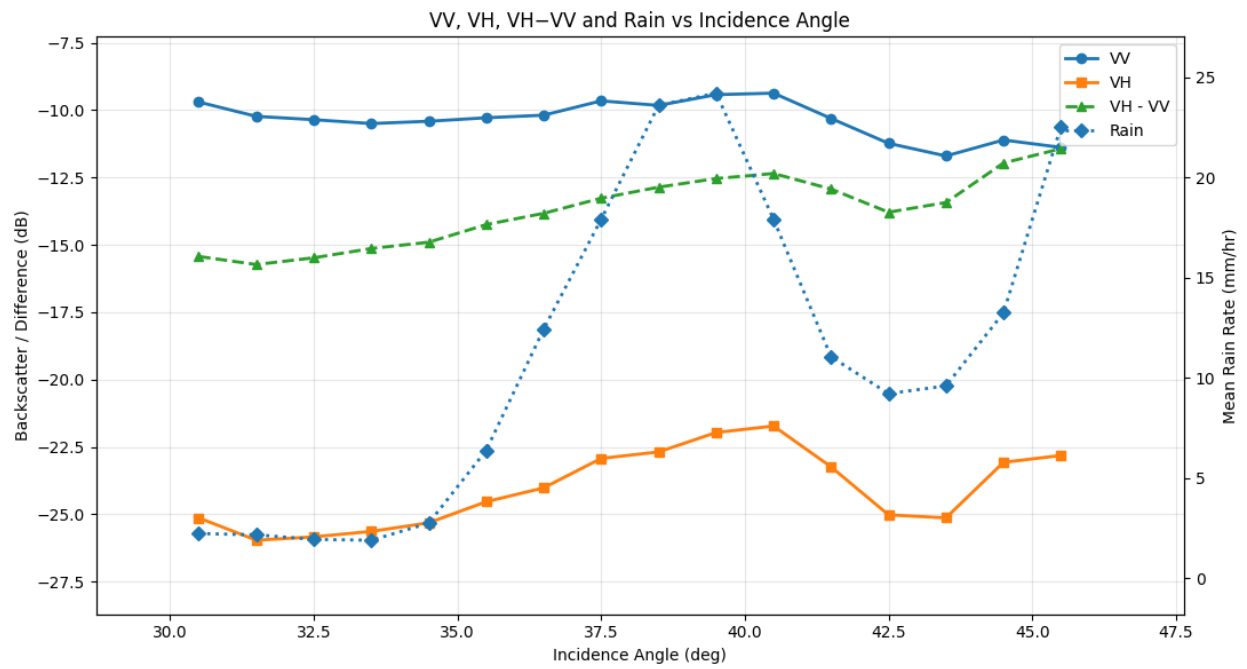


Incidence Angle Dependence Analysis

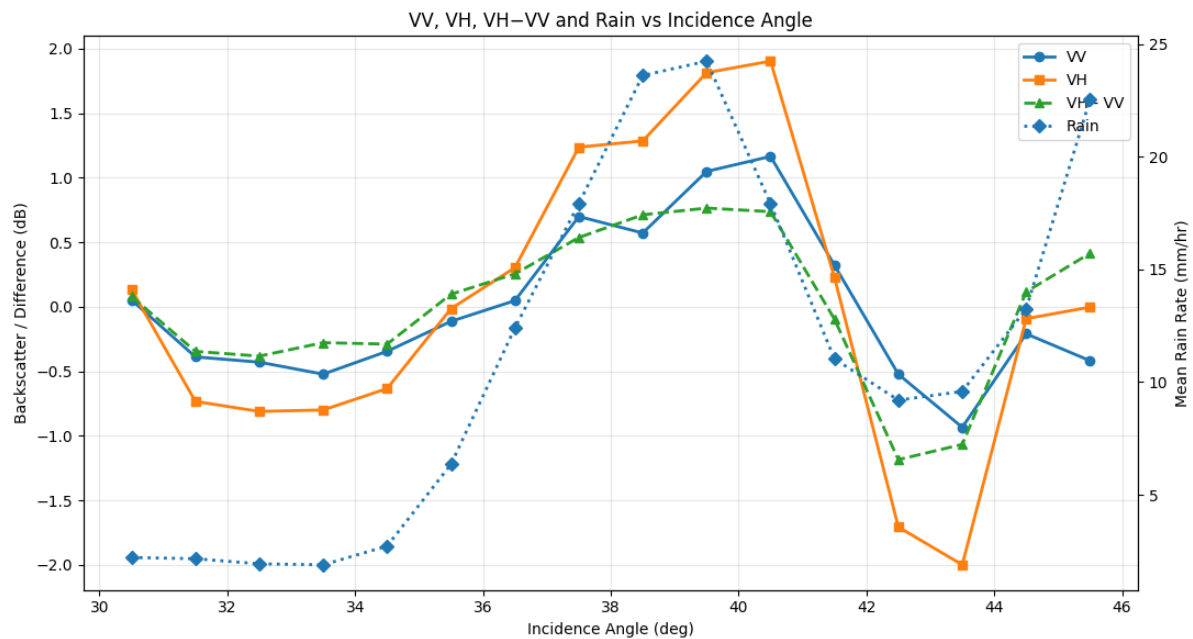
The incidence angle is known to influence SAR backscatter measurements, particularly over ocean surfaces where variations in viewing geometry can alter the observed scattering characteristics. To investigate this effect, scatter plots of VV, VH, and polarization-derived quantities against the extracted local incidence angle were generated. These visualizations enabled the identification of systematic angular trends and assessment of the extent to which incidence angle variations contribute to backscatter variability.

To further isolate rainfall-induced signatures from geometric effects, a simple linear regression model was employed to characterize the dependence of backscatter on incidence angle. The fitted regression relationship was subsequently used to estimate and remove the incidence angle component from the observations, yielding residual backscatter values that are expected to be less affected by acquisition geometry. Comparison of the original and corrected datasets provides insight into the significance of incidence angle normalization and its potential role in improving the characterization of rainfall-related scattering mechanisms in Sentinel-1 SAR imagery.

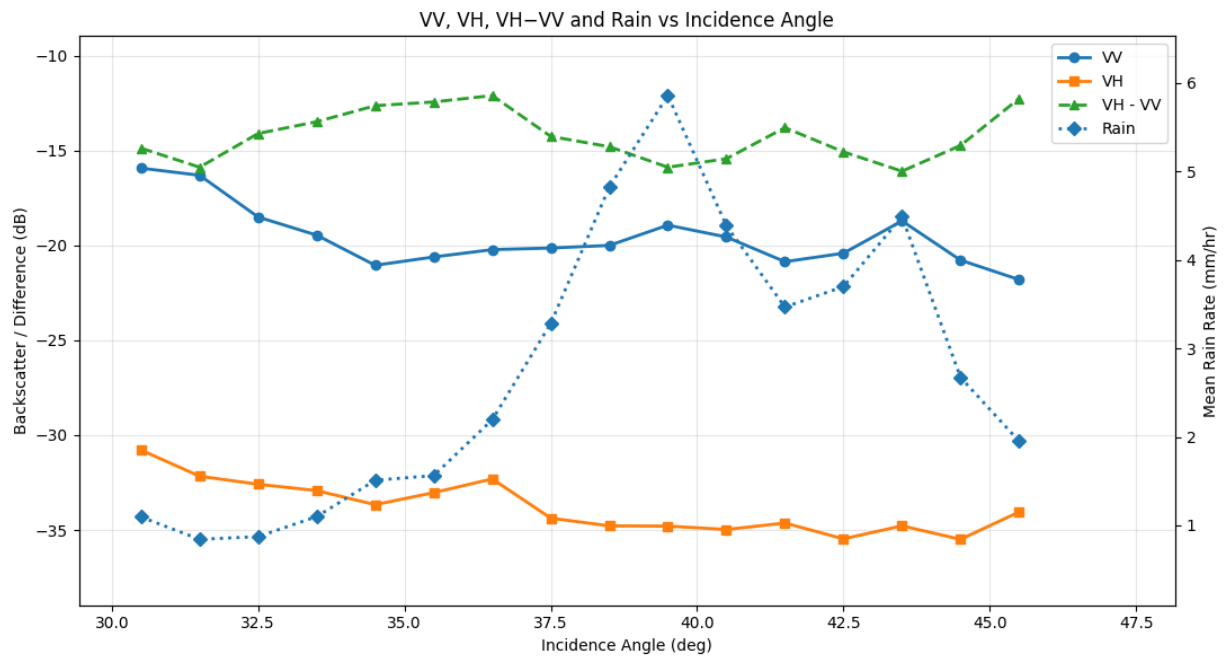
For Cyclone



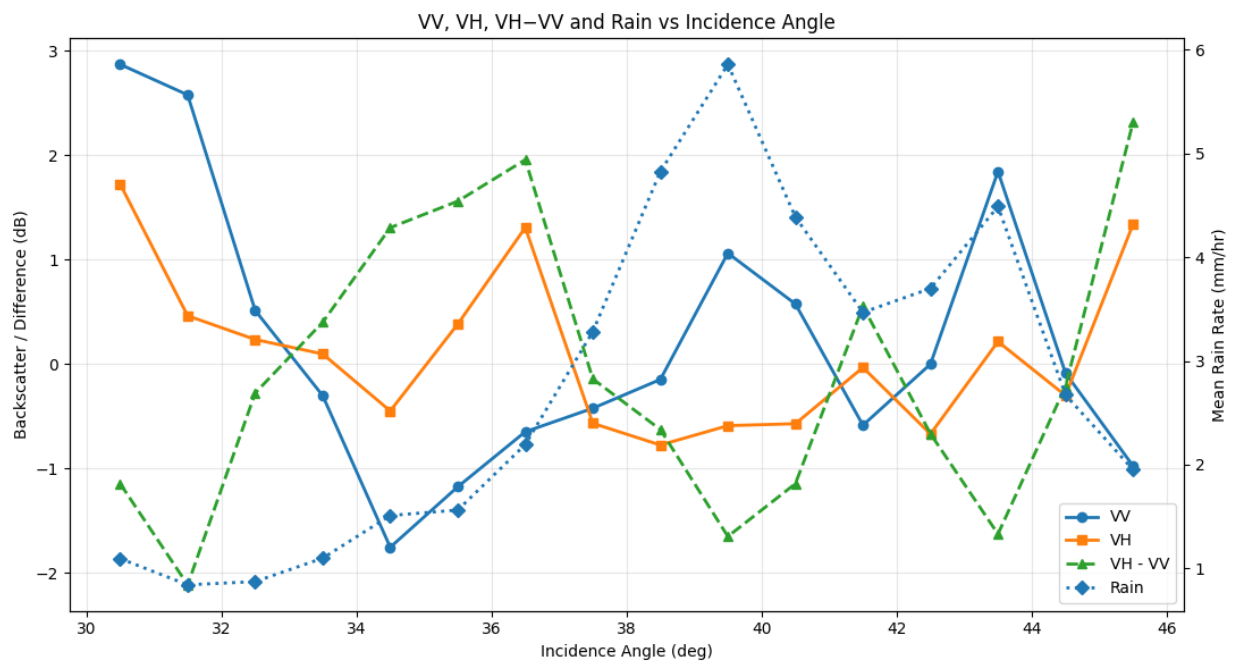
For Cyclone after Correction



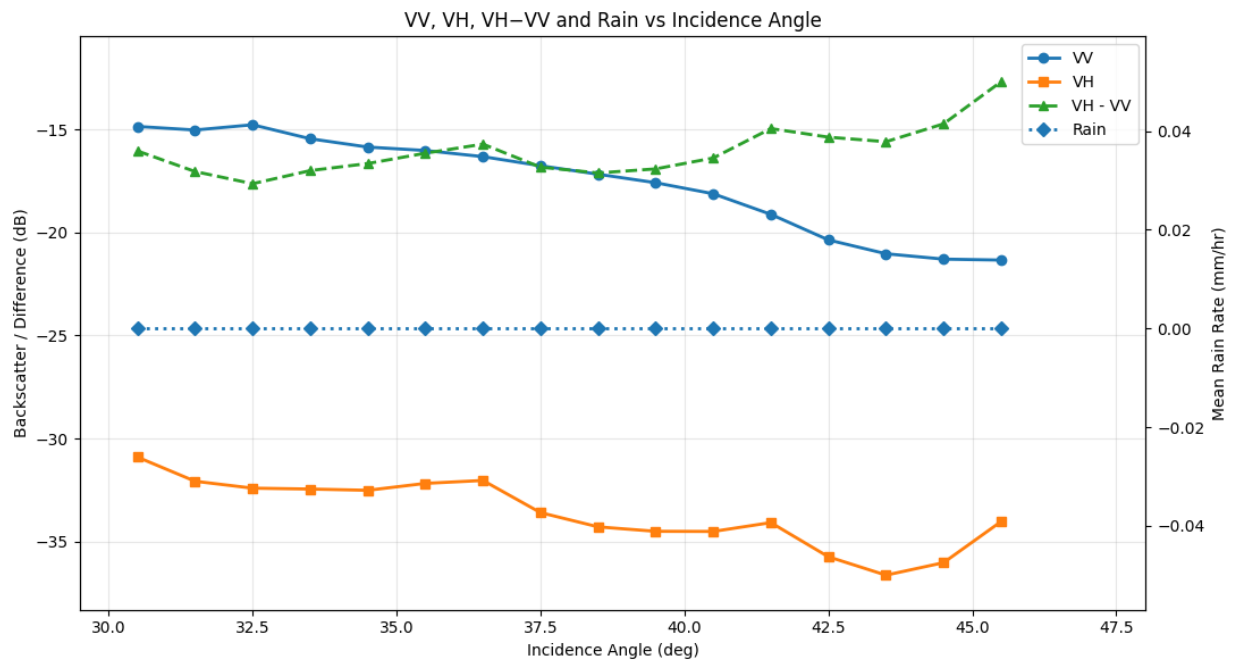
For Moderate Rain



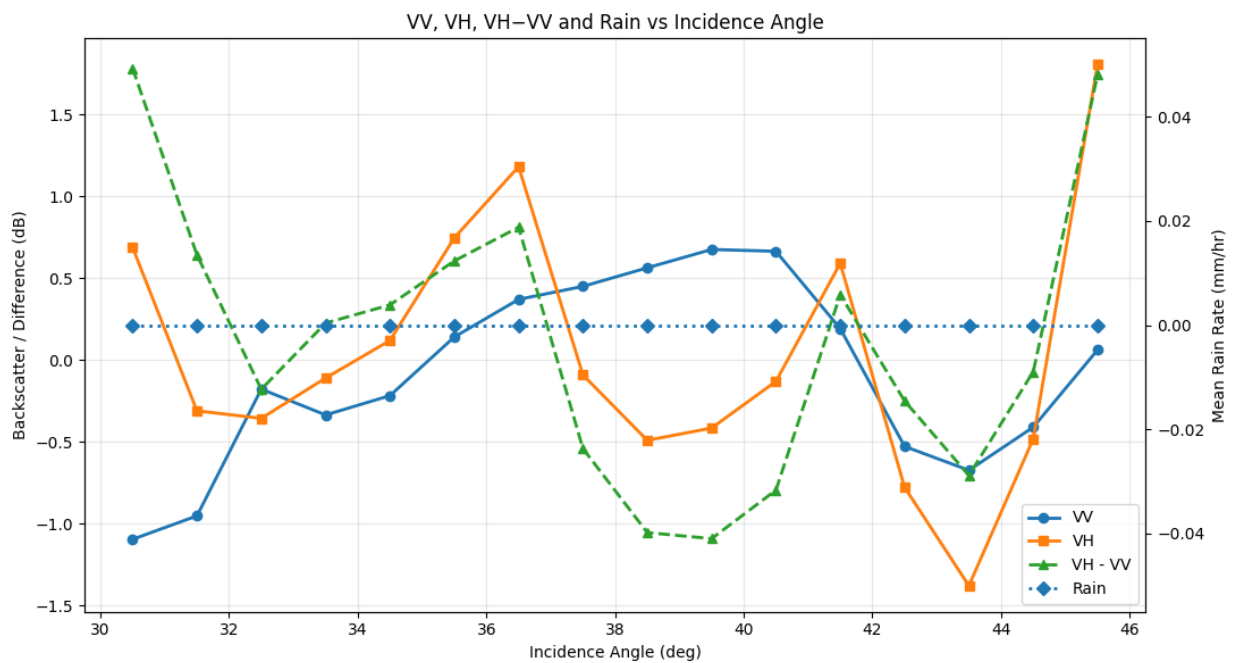
For Moderate rain after correction



For Same area in dry conditions



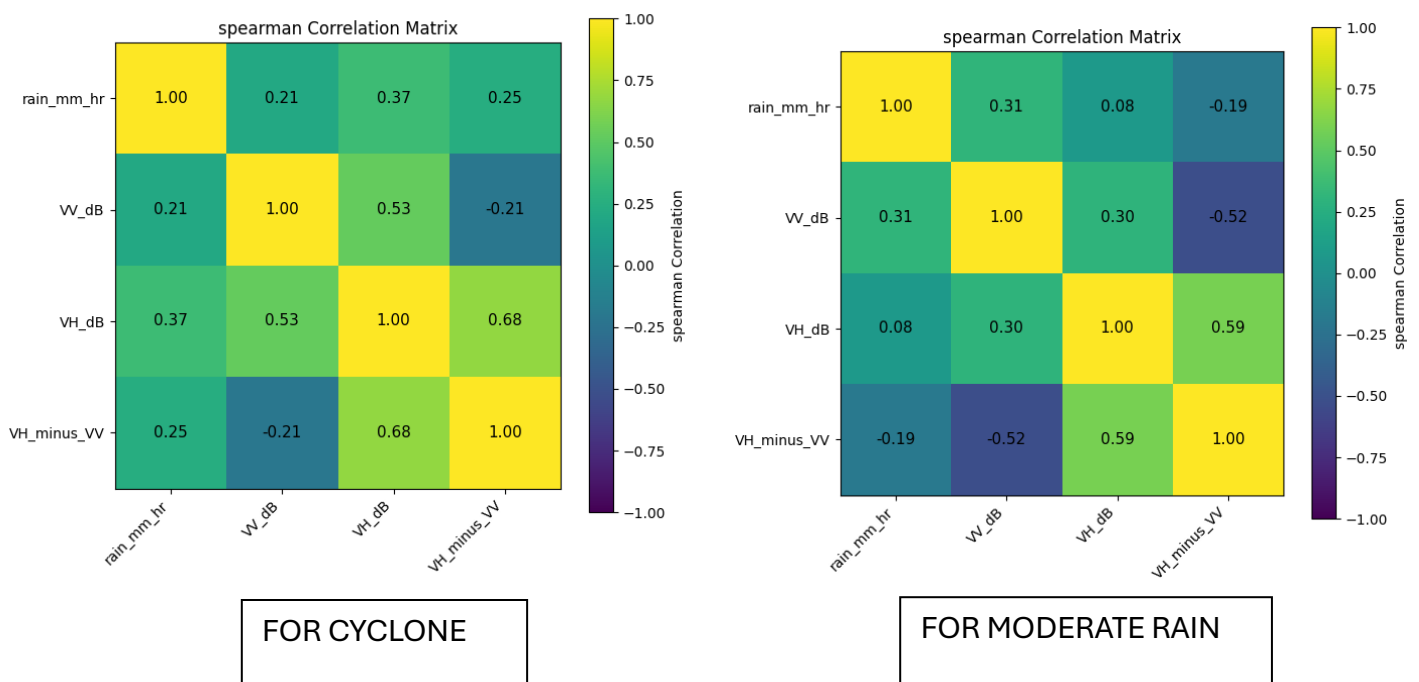
For Same area in dry conditions after correction



Rainfall–Backscatter Relationship Analysis

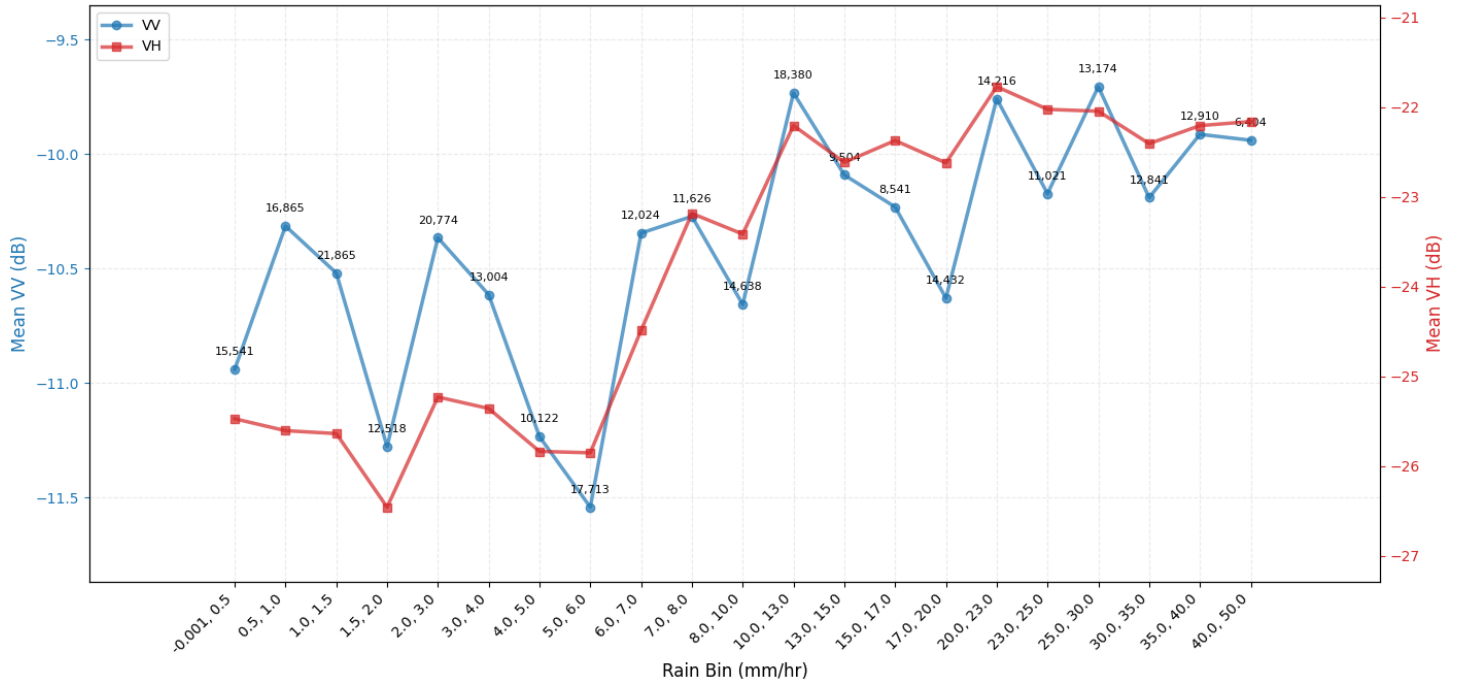
To investigate the sensitivity of Sentinel-1 SAR observations to precipitation, the collocated rainfall estimates from IMERG were compared with the corresponding **VV** and **VH** backscatter measurements. Scatter plots, rain-bin analyses, and threshold-based statistics were employed to examine variations in backscatter with increasing rainfall intensity. Correlation coefficients, including Pearson and Spearman metrics, were computed to quantify the strength and monotonicity of the observed relationships.

The analysis was performed separately for **VV**, **VH**, and polarization-derived quantities such as the **VH–VV difference**, enabling a comparative assessment of their respective sensitivities to rainfall. Particular attention was given to identifying systematic trends in backscatter under rainy and non-rainy conditions, evaluating the potential of cross-polarized observations for precipitation characterization, and distinguishing rainfall-induced signatures from other sources of variability. These investigations provide insight into the feasibility of utilizing Sentinel-1 SAR observations for qualitative and quantitative rainfall studies over oceanic environments.

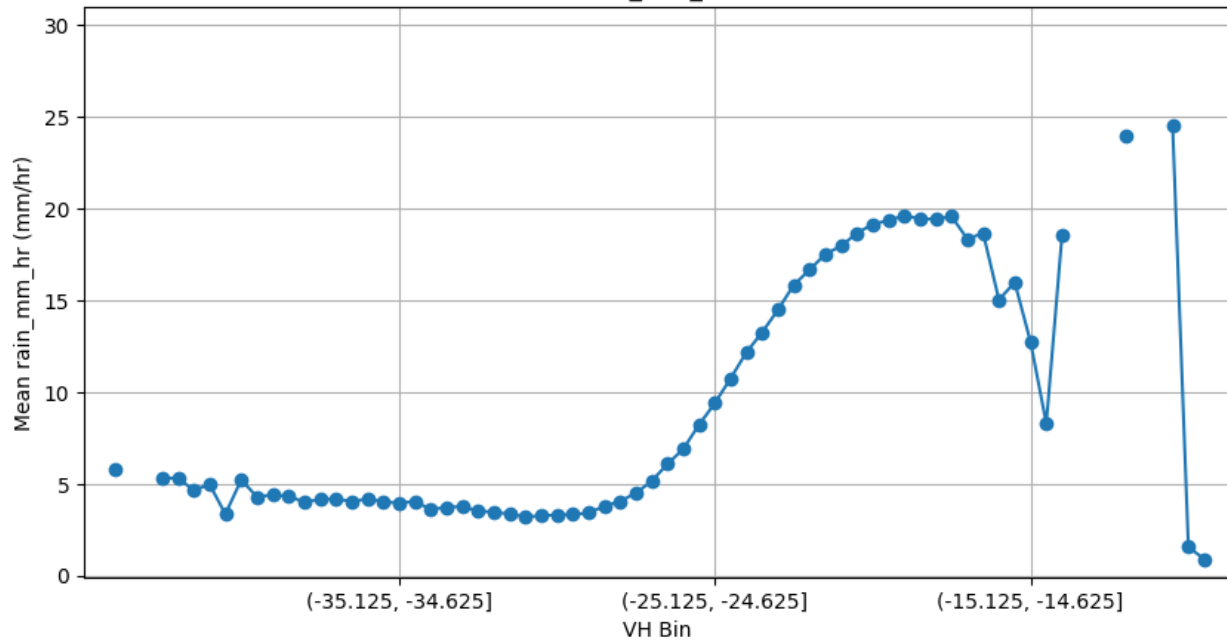


For Cyclone

Mean VV and VH Response to Rainfall

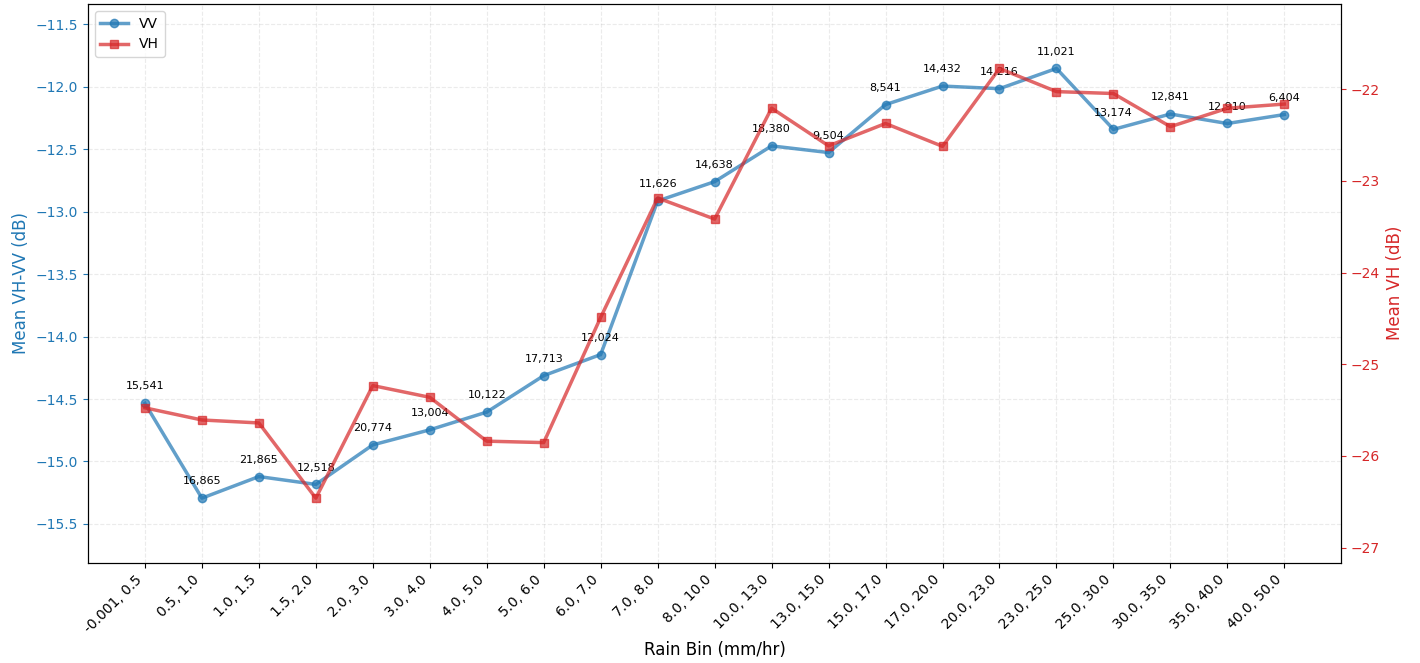


Mean rain_mm_hr vs VH

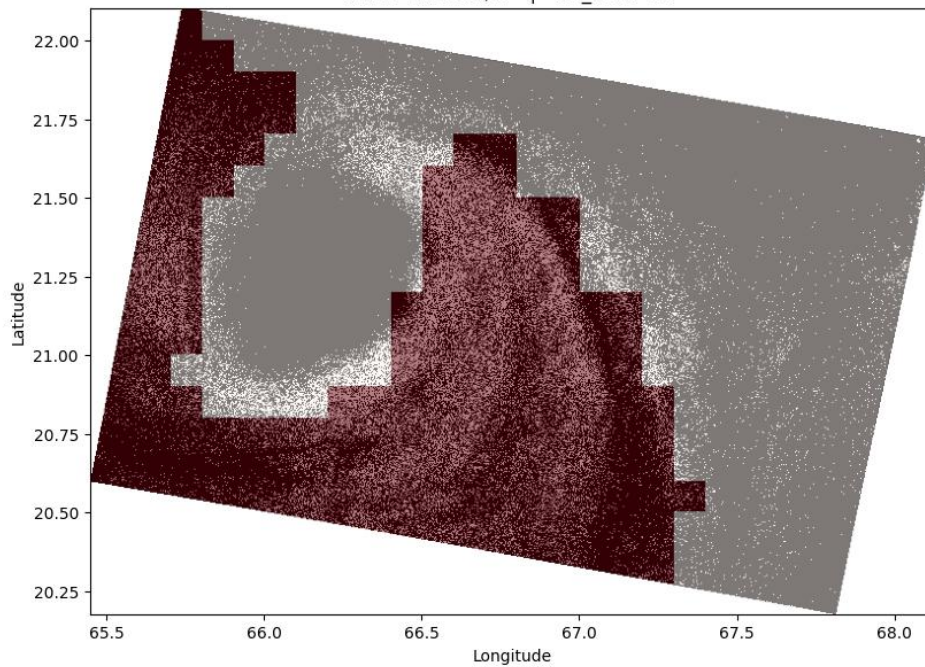


For Cyclone

Mean VH-VV(dB) and VH Response to Rainfall



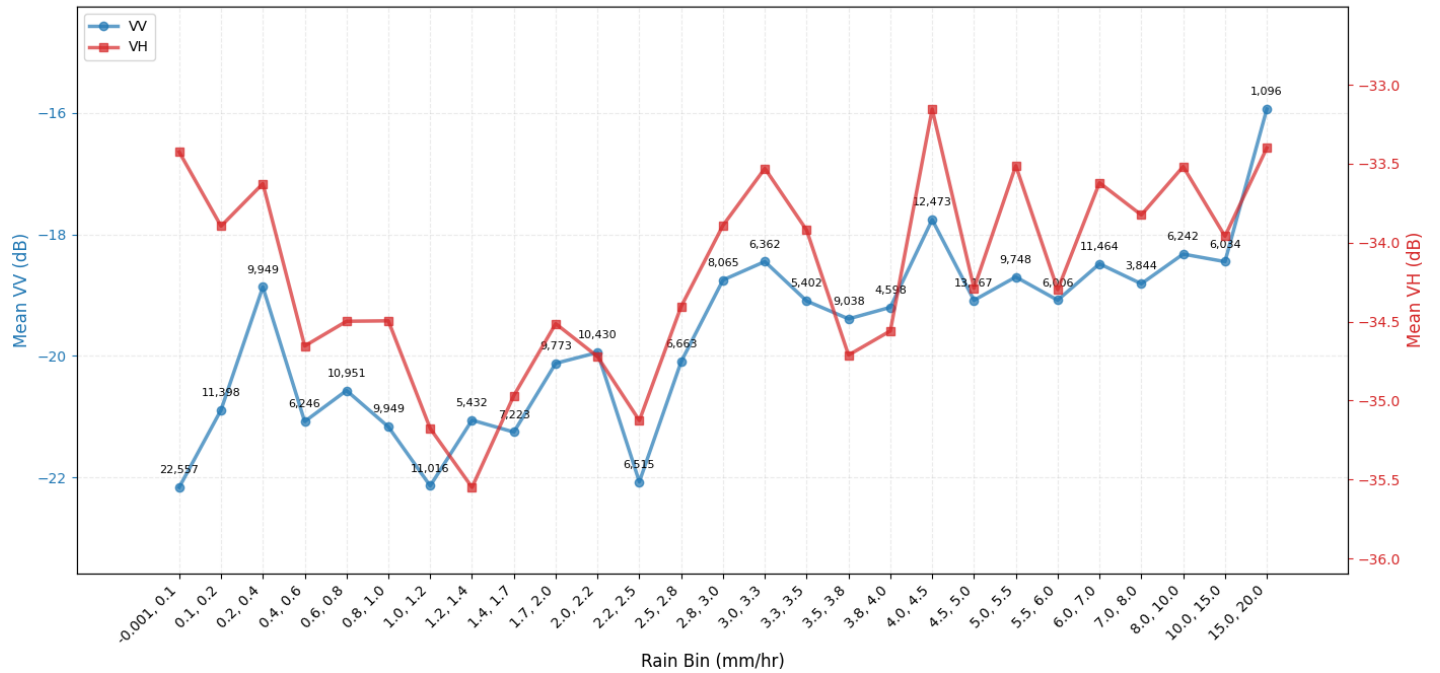
Rain ≥ 10 mm/hr | VH_{dB} ≥ -21



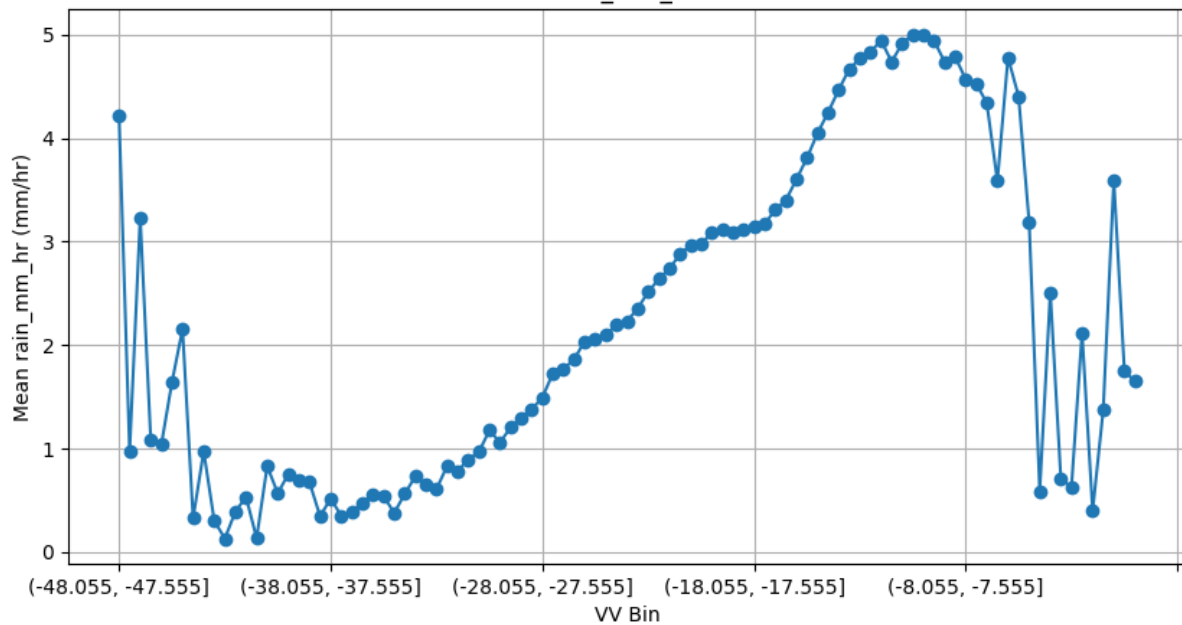
Rain Transparency 0.5

For Moderate rain

Mean VV and VH Response to Rainfall

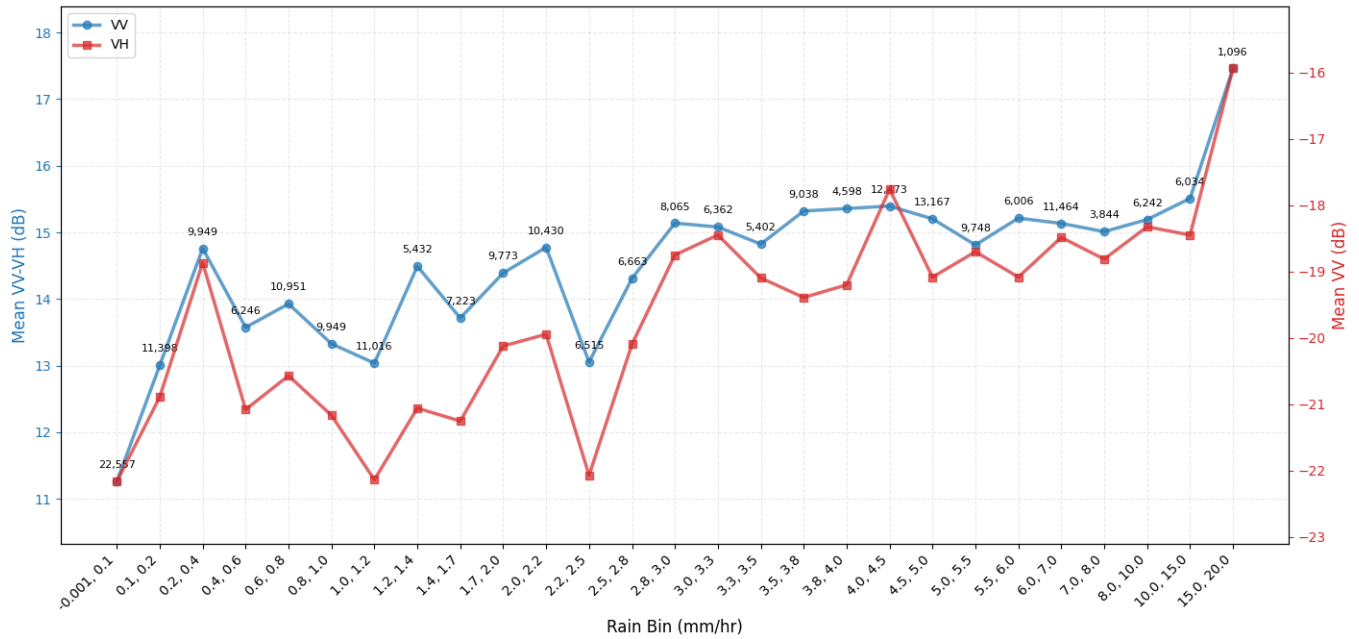


Mean rain_mm_hr vs VV

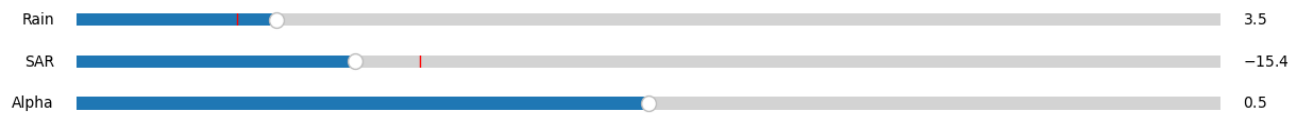
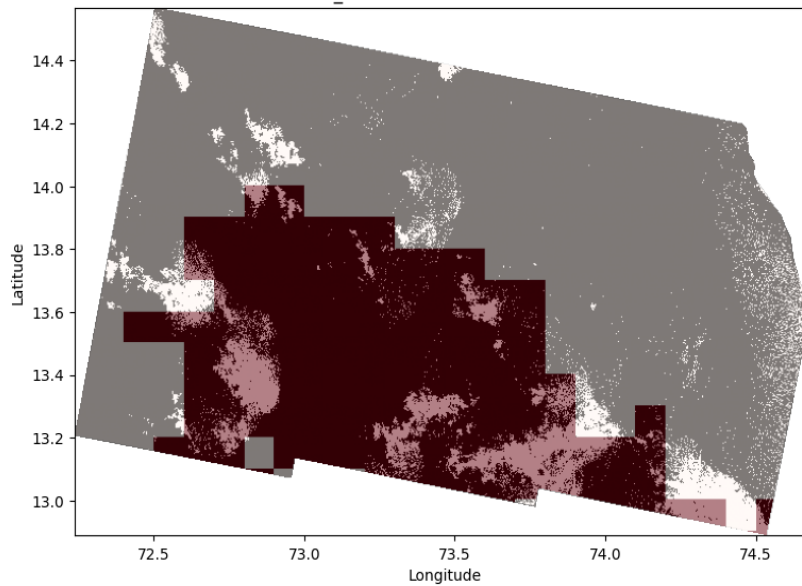


For Moderate rain

Mean VV-VH(dB) and VV Response to Rainfall



VV_dB $\geq$ -15.40 Rain $\geq$ 3.50



#Conclusion

This work presents a framework for generating collocated Sentinel-1 SAR and GPM IMERG datasets to study the influence of rainfall on SAR backscatter over oceanic regions. The developed pipeline integrates SAR preprocessing, IMERG extraction, spatial collocation, land masking, exploratory analysis, and visualization into a flexible workflow that can be applied to different Sentinel-1 scenes and weather events.

The use of NetCDF and Parquet formats enabled efficient storage and processing of multidimensional SAR observations and large collocated datasets, while stride-based sampling and KDTree-based nearest-neighbor matching helped maintain computational efficiency. Additional analyses, including IMERG-SAR overlays, incidence angle dependence studies, and simple incidence angle correction using linear regression, provided further insight into the factors affecting SAR backscatter measurements.

Preliminary investigations suggest that rainfall produces observable changes in Sentinel-1 backscatter, particularly in the cross-polarized VH channel during heavy rain, and VV during moderate, although the relationship is influenced by factors such as incidence angle, ocean state, and the relatively coarse spatial resolution of IMERG. Nevertheless, the developed framework provides a useful foundation for further studies on rainfall characterization, larger multi-scene analyses, and the potential application of machine learning techniques for precipitation retrieval from SAR observations.

Overall, this study demonstrates that combining high-resolution SAR observations with satellite-derived precipitation products is a promising approach for investigating rainfall signatures over marine environments and establishes a reusable framework that can be extended and refined for future research.



Thank
You!